

Integration Theory MATM39: Summary of Important Results

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Chapter 1

Sigma Algebras

Definition 1. Let X be an arbitrary set. A collection $\mathcal{A}$ of subsets of X is called a σ -algebra on X if:

- (a) $X \in \mathcal{A}$,
- (b) if $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$,
- (c) if $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$,
- (d) if $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$, then $\bigcap_{i=1}^{\infty} A_i \in \mathcal{A}$.

Proof. Assume (a), (b), and (c). Let $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$. Then $A_i^c \in \mathcal{A}$ for all i , and by (c), $\bigcup_{i=1}^{\infty} A_i^c \in \mathcal{A}$. By (b) again, $(\bigcup_{i=1}^{\infty} A_i^c)^c \in \mathcal{A}$. By De Morgan's law,

$$\bigcap_{i=1}^{\infty} A_i = \left(\bigcup_{i=1}^{\infty} A_i^c \right)^c \in \mathcal{A}$$

□

Proposition 1. Let X be a set. Then the intersection of an arbitrary nonempty collection of σ -algebras on X is a σ -algebra on X .

Proof. Let $\mathcal{C}$ be a nonempty collection of σ -algebras on X , and define

$$\mathcal{A} := \bigcap_{\mathcal{B} \in \mathcal{C}} \mathcal{B}.$$

Since every $\mathcal{B} \in \mathcal{C}$ contains X , we have $X \in \mathcal{A}$. If $A \in \mathcal{A}$, then $A \in \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$. Since each $\mathcal{B}$ is a σ -algebra, $A^c \in \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$, hence $A^c \in \mathcal{A}$. If $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$, then $\{A_i\}_{i=1}^{\infty} \subset \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$. Hence

$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{B}$$

for every $\mathcal{B} \in \mathcal{C}$, and therefore

$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}.$$

Thus $\mathcal{A}$ is a σ -algebra on X .

□

Corollary 1. *Let X be a set, and let $\mathcal{F}$ be a family of subsets of X . Then there exists a smallest σ -algebra on X that contains $\mathcal{F}$.*

Proof. Let $\mathcal{C}$ be the collection of all σ -algebras on X that contain $\mathcal{F}$. This collection is nonempty, since $\mathcal{P}(X)$ is a σ -algebra on X and contains $\mathcal{F}$.

Define

$$\sigma(\mathcal{F}) := \bigcap_{\mathcal{A} \in \mathcal{C}} \mathcal{A}.$$

By the previous proposition, $\sigma(\mathcal{F})$ is a σ -algebra on X . It contains $\mathcal{F}$, since every $\mathcal{A} \in \mathcal{C}$ contains $\mathcal{F}$. Moreover, if $\mathcal{B}$ is any σ -algebra on X that contains $\mathcal{F}$, then $\mathcal{B} \in \mathcal{C}$, and therefore

$$\sigma(\mathcal{F}) \subset \mathcal{B}.$$

Thus $\sigma(\mathcal{F})$ is the smallest σ -algebra on X containing $\mathcal{F}$. □

Definition 2. *The Borel σ -algebra on $\mathbb{R}^d$ is the σ -algebra generated by the open subsets of $\mathbb{R}^d$. It is denoted by $\mathcal{B}(\mathbb{R}^d)$. Its elements are called Borel sets.*

Proposition 2. *The Borel σ -algebra $\mathcal{B}(\mathbb{R})$ is generated by each of the following collections:*

- (a) *the collection of all closed subsets of $\mathbb{R}$,*
- (b) *the collection of all intervals of the form $(-\infty, b]$,*
- (c) *the collection of all intervals of the form $(a, b]$.*

Proof. Let $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$ denote the σ -algebras generated by the collections in (a), (b), and (c), respectively.

Since $\mathcal{B}(\mathbb{R})$ contains all open sets and is closed under complements, it contains all closed sets. Hence

$$\mathcal{B}(\mathbb{R}) \supset \mathcal{B}_1.$$

Every set $(-\infty, b]$ is closed, so $\mathcal{B}_1 \supset \mathcal{B}_2$. Moreover,

$$(a, b] = (-\infty, b] \cap (-\infty, a]^c,$$

so $\mathcal{B}_2 \supset \mathcal{B}_3$.

Finally, let $G \subset \mathbb{R}$ be open. For each $x \in G$, since G is open, there exist rational numbers $a, b \in \mathbb{Q}$ such that

$$x \in (a, b] \subset G.$$

Hence

$$G = \bigcup_{\substack{a, b \in \mathbb{Q} \\ (a, b] \subset G}} (a, b].$$

This is a countable union, since $\mathbb{Q}^2$ is countable. Therefore $G \in \mathcal{B}_3$. Thus every open set belongs to $\mathcal{B}_3$, so

$$\mathcal{B}(\mathbb{R}) \subset \mathcal{B}_3.$$

Therefore all four σ -algebras are equal. □

Measures

Definition 3. Let X be a set and $\mathcal{A}$ a σ -algebra on X . A function $\mu : \mathcal{A} \rightarrow [0, \infty]$ is called a *measure* if

- (a) $\mu(\emptyset) = 0$,
- (b) for every $\{A_i\}_{i=1}^{\infty}$ of pairwise disjoint sets in $\mathcal{A}$, $\mu(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$.

Proposition 3. Let $(X, \mathcal{A}, \mu)$ be a measure space.

- (a) If $\{A_k\}$ is an increasing sequence in $\mathcal{A}$, then

$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \lim_{k \rightarrow \infty} \mu(A_k).$$

- (b) If $\{A_k\}$ is a decreasing sequence in $\mathcal{A}$ and $\mu(A_n) < \infty$ for some n , then

$$\mu\left(\bigcap_{k=1}^{\infty} A_k\right) = \lim_{k \rightarrow \infty} \mu(A_k).$$

Proof. (a) Define $B_1 = A_1$ and $B_k = A_k \setminus A_{k-1}$ for $k \geq 2$. Then $\{B_k\}$ are disjoint and $\bigcup_k A_k = \bigcup_k B_k$. Hence

$$\mu\left(\bigcup_k A_k\right) = \sum_k \mu(B_k) = \lim_{k \rightarrow \infty} \sum_{i=1}^k \mu(B_i) = \lim_{k \rightarrow \infty} \mu(A_k).$$

(b) Since $\mu(A_n) < \infty$ for some n , we may replace $\{A_k\}$ by the tail $\{A_{k+n-1}\}$ and assume $\mu(A_1) < \infty$. Let $C_k = A_1 \setminus A_k$. Then $\{C_k\}$ is increasing and

$$\bigcup_k C_k = A_1 \setminus \bigcap_k A_k.$$

By (a),

$$\mu(A_1 \setminus \bigcap_k A_k) = \lim_{k \rightarrow \infty} \mu(C_k) = \lim_{k \rightarrow \infty} (\mu(A_1) - \mu(A_k)),$$

which gives $\mu(\bigcap_k A_k) = \lim_k \mu(A_k)$. □

Outer Measures

Definition 4. Let X be a set. An *outer measure* on X is a function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ such that

- (a) $\mu^*(\emptyset) = 0$,
- (b) if $A \subset B$, then $\mu^*(A) \leq \mu^*(B)$,
- (c) for any sequence $\{A_n\}$ of subsets of X , $\mu^*(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$.

Definition 5. The Lebesgue outer measure on $\mathbb{R}$ is the function $\lambda^* : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$ defined by

$$\lambda^*(A) = \inf \left\{ \sum_{i=1}^{\infty} (b_i - a_i) : A \subset \bigcup_{i=1}^{\infty} (a_i, b_i) \right\}.$$

Definition 6. The Lebesgue outer measure on $\mathbb{R}^d$ is defined by

$$\lambda^*(A) = \inf \left\{ \sum_{i=1}^{\infty} \text{vol}(R_i) : A \subset \bigcup_{i=1}^{\infty} R_i \right\},$$

where each R_i is a bounded open rectangle in $\mathbb{R}^d$ and

$$\text{vol}(I_1 \times \cdots \times I_d) = \prod_{j=1}^d |I_j|.$$

Definition 7. Let μ^* be an outer measure on X . A set $B \subset X$ is called μ^* -measurable if for all $A \subset X$,

$$\mu^*(A) = \mu^*(A \cap B) + \mu^*(A \cap B^c).$$

Remark 1. Outer measures are defined on all subsets of X , but are only subadditive in general. Carathéodory's criterion selects a class of sets on which μ^* becomes additive. This allows us to restrict μ^* to a σ -algebra and obtain a genuine measure (in particular, the Lebesgue measure).

Theorem 1. Let X be a set and μ^* an outer measure on X . Let

$$\mathcal{M}_{\mu^*} := \{B \subset X : B \text{ is } \mu^*\text{-measurable}\}.$$

Then:

- (a) $\mathcal{M}_{\mu^*}$ is a σ -algebra,
- (b) the restriction of μ^* to $\mathcal{M}_{\mu^*}$ is a measure.

Proof sketch. One shows that $\mathcal{M}_{\mu^*}$ is closed under complements directly from the definition. Closure under finite unions follows by applying the defining identity to B_1 and B_2 and decomposing sets appropriately.

Closure under countable unions is obtained by first proving the result for finite disjoint unions and then passing to the limit using subadditivity of μ^* .

Finally, countable additivity of μ^* on $\mathcal{M}_{\mu^*}$ follows by applying the defining property to disjoint measurable sets and using the previous decomposition. $\square$

Remark 2. Applying this theorem to the Lebesgue outer measure λ^* on $\mathbb{R}^d$ produces a σ -algebra $\mathcal{M}_{\lambda^*}$, called the Lebesgue σ -algebra. The restriction of λ^* to this σ -algebra is the Lebesgue measure. Thus, outer measure together with Carathéodory's criterion provides a systematic way to construct Lebesgue measure.

Example 1 (Cantor set). Let $K_0 = [0, 1]$. Construct K_n by removing the open middle third of each interval in K_{n-1} . The Cantor set is $K = \bigcap_{n=0}^{\infty} K_n$.

Proposition 4. The Cantor set K is compact, uncountable, and has Lebesgue measure zero.

Idea. Each K_n is a finite union of closed intervals, hence compact, and $K = \bigcap_n K_n$ is compact. One shows K has the cardinality of the continuum via ternary expansions.

Moreover, $\lambda(K_n) = (2/3)^n$, so

$$\lambda(K) \leq \lambda(K_n) = (2/3)^n \rightarrow 0,$$

hence $\lambda(K) = 0$. □

Theorem 2. *There exists a subset of $(0, 1)$ that is not Lebesgue measurable.*

Idea. Define $x \sim y$ if $x - y \in \mathbb{Q}$ and choose one representative from each equivalence class (using the axiom of choice) to form a set $E \subset (0, 1)$.

Consider the translates $E + r_n$ by rational numbers r_n . These sets are disjoint and their union covers $(0, 1)$ up to inclusion. Translation invariance and countable additivity would force a contradiction, so E cannot be measurable. □

Remark 3. *Not every subset of $\mathbb{R}$ can be assigned a consistent measure: the existence of non-measurable sets shows that countable additivity and translation invariance cannot hold on all subsets.*

Therefore, one must restrict attention to a σ -algebra (such as the Lebesgue σ -algebra) on which a well-defined measure exists.

Chapter 2

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Definition 8. *Let $(X, \mathcal{A}, \mu)$ be a measure space. A nonnegative simple measurable function is a function of the form*

$$f = \sum_{i=1}^n a_i \mathbf{1}_{A_i},$$

where $a_i \geq 0$ and $A_i \in \mathcal{A}$ are disjoint. Its integral is defined by

$$\int f \, d\mu := \sum_{i=1}^n a_i \mu(A_i).$$

Definition 9. *Let $f : X \rightarrow [0, \infty]$ be measurable. The integral of f is defined by*

$$\int f \, d\mu := \sup \left\{ \int g \, d\mu : g \in \mathcal{S}_+, g \leq f \right\},$$

where $\mathcal{S}_+$ denotes the set of nonnegative simple measurable functions.

Proposition 5. *Every nonnegative measurable function $f : X \rightarrow [0, \infty]$ is the pointwise limit of an increasing sequence of nonnegative simple measurable functions.*

Reference. This is Proposition 2.1.8 in Cohn. It guarantees that nonnegative measurable functions can be approximated from below by simple functions. □

Proposition 6. *If $f_n \in \mathcal{S}_+$, $f_n \uparrow f$, and f is nonnegative measurable, then*

$$\int f \, d\mu = \lim_{n \rightarrow \infty} \int f_n \, d\mu.$$

Reference. This is Proposition 2.3.3 in Cohn. It shows that the supremum definition agrees with taking limits along increasing simple approximations. Proposition 2.3.2 is the corresponding result when the limiting function is itself simple. $\square$

Remark 4. *Thus the integral is first defined directly for nonnegative simple functions. For arbitrary nonnegative measurable functions, it is defined as the supremum of the integrals of all simple functions below it. Propositions 2.1.8 and 2.3.3 show that this construction is compatible with approximation from below.*

Definition 10. *Let $f : X \rightarrow [-\infty, \infty]$ be measurable. Define*

$$f^+ := \max(f, 0), \quad f^- := \max(-f, 0).$$

Then

$$f = f^+ - f^-, \quad |f| = f^+ + f^-.$$

If at least one of $\int f^+ d\mu$ and $\int f^- d\mu$ is finite, then $\int f d\mu$ is said to exist, and is defined by

$$\int f d\mu := \int f^+ d\mu - \int f^- d\mu.$$

If both $\int f^+ d\mu$ and $\int f^- d\mu$ are finite, then f is called integrable.

Definition 11. *If $A \in \mathcal{A}$, then f is called integrable over A if $f\mathbf{1}_A$ is integrable. In that case,*

$$\int_A f d\mu := \int f\mathbf{1}_A d\mu.$$

Proposition 7 (Proposition 2.3.2). *Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $f \in \mathcal{S}_+$, and let $\{f_n\}$ be an increasing sequence in $\mathcal{S}_+$ such that*

$$f = \lim_{n \rightarrow \infty} f_n$$

pointwise on X . Then

$$\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu.$$

Proof. By monotonicity,

$$\int f_1 d\mu \leq \int f_2 d\mu \leq \cdots \leq \int f d\mu,$$

so

$$\lim_{n \rightarrow \infty} \int f_n d\mu \leq \int f d\mu.$$

We prove the reverse inequality. Let $0 < \varepsilon < 1$. Write

$$f = \sum_{i=1}^k a_i \mathbf{1}_{A_i},$$

where $a_i > 0$ and the sets $A_i \in \mathcal{A}$ are disjoint. For each n and i , define

$$A(n, i) := \{x \in A_i : f_n(x) \geq (1 - \varepsilon)a_i\}.$$

Then $A(n, i) \in \mathcal{A}$, the sequence $\{A(n, i)\}_{n=1}^{\infty}$ is increasing, and

$$A_i = \bigcup_{n=1}^{\infty} A(n, i).$$

Define

$$g_n := \sum_{i=1}^k (1 - \varepsilon) a_i \mathbf{1}_{A(n, i)}.$$

Then $g_n \in \mathcal{S}_+$ and $g_n \leq f_n$. Hence

$$\int g_n d\mu \leq \int f_n d\mu.$$

By continuity from below of measures,

$$\lim_{n \rightarrow \infty} \int g_n d\mu = \lim_{n \rightarrow \infty} \sum_{i=1}^k (1 - \varepsilon) a_i \mu(A(n, i)) = \sum_{i=1}^k (1 - \varepsilon) a_i \mu(A_i) = (1 - \varepsilon) \int f d\mu.$$

Therefore

$$(1 - \varepsilon) \int f d\mu \leq \lim_{n \rightarrow \infty} \int f_n d\mu.$$

Since $\varepsilon > 0$ was arbitrary,

$$\int f d\mu \leq \lim_{n \rightarrow \infty} \int f_n d\mu.$$

Thus

$$\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu.$$

□

Proposition 8 (Proposition 2.3.3). *Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $f : X \rightarrow [0, \infty]$ be measurable, and let $\{f_n\}$ be an increasing sequence of functions in $\mathcal{S}_+$ such that*

$$f = \lim_{n \rightarrow \infty} f_n$$

pointwise on X . Then

$$\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu.$$

Proof. By monotonicity of the integral,

$$\int f_1 d\mu \leq \int f_2 d\mu \leq \cdots \leq \int f d\mu.$$

Hence

$$\lim_{n \rightarrow \infty} \int f_n d\mu \leq \int f d\mu.$$

For the reverse inequality, let $g \in \mathcal{S}_+$ with $g \leq f$. Then $\{g \wedge f_n\}$ is an increasing sequence in $\mathcal{S}_+$ and

$$g = \lim_{n \rightarrow \infty} (g \wedge f_n).$$

By Proposition 2.3.2,

$$\int g \, d\mu = \lim_{n \rightarrow \infty} \int (g \wedge f_n) \, d\mu.$$

Since $g \wedge f_n \leq f_n$, we get

$$\int g \, d\mu \leq \lim_{n \rightarrow \infty} \int f_n \, d\mu.$$

Taking the supremum over all $g \in \mathcal{S}_+$ with $g \leq f$ gives

$$\int f \, d\mu \leq \lim_{n \rightarrow \infty} \int f_n \, d\mu.$$

Therefore,

$$\int f \, d\mu = \lim_{n \rightarrow \infty} \int f_n \, d\mu.$$

□

Proposition 9 (Proposition 2.3.9). *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f, g : X \rightarrow [-\infty, \infty]$ be measurable functions such that $f = g$ almost everywhere. If either $\int f \, d\mu$ or $\int g \, d\mu$ exists, then both exist and*

$$\int f \, d\mu = \int g \, d\mu.$$

Proof. First assume that f and g are nonnegative. Let

$$A := \{x \in X : f(x) \neq g(x)\}.$$

Then $\mu(A) = 0$. Define

$$h(x) = \begin{cases} +\infty, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Then $\int h \, d\mu = 0$. Since $f \leq g + h$, monotonicity gives

$$\int f \, d\mu \leq \int g \, d\mu + \int h \, d\mu = \int g \, d\mu.$$

Similarly, $g \leq f + h$, so

$$\int g \, d\mu \leq \int f \, d\mu.$$

Thus $\int f \, d\mu = \int g \, d\mu$.

For general f and g , apply the nonnegative case to f^+ and g^+ , and to f^- and g^- . Hence the integrals exist together and are equal. □

Theorem 3 (Monotone Convergence Theorem). *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f, f_1, f_2, \dots$ be $[0, \infty]$ -valued measurable functions on X . Suppose that*

$$f_1 \leq f_2 \leq \dots$$

and

$$f = \lim_{n \rightarrow \infty} f_n$$

hold almost everywhere. Then

$$\int f \, d\mu = \lim_{n \rightarrow \infty} \int f_n \, d\mu.$$

Proof. First assume that $f_n \uparrow f$ pointwise everywhere. By monotonicity of the integral,

$$\int f_1 d\mu \leq \int f_2 d\mu \leq \cdots \leq \int f d\mu.$$

Hence

$$\lim_{n \rightarrow \infty} \int f_n d\mu \leq \int f d\mu.$$

We prove the reverse inequality. For each n , choose an increasing sequence $\{g_{n,k}\}_{k=1}^{\infty}$ of nonnegative simple functions such that

$$g_{n,k} \uparrow f_n.$$

This is possible by Proposition 2.1.8. Define

$$h_n := \max(g_{1,n}, g_{2,n}, \dots, g_{n,n}).$$

Then $\{h_n\}$ is an increasing sequence of nonnegative simple functions, $h_n \leq f_n$, and

$$h_n \uparrow f.$$

Therefore, by Proposition 2.3.3,

$$\int f d\mu = \lim_{n \rightarrow \infty} \int h_n d\mu.$$

Since $h_n \leq f_n$, monotonicity gives

$$\int h_n d\mu \leq \int f_n d\mu.$$

Thus

$$\int f d\mu = \lim_{n \rightarrow \infty} \int h_n d\mu \leq \lim_{n \rightarrow \infty} \int f_n d\mu.$$

Combining the two inequalities gives

$$\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu.$$

If the assumptions hold only almost everywhere, choose a null set N outside which they hold. Apply the first part to $f_n \mathbf{1}_{N^c}$ and $f \mathbf{1}_{N^c}$. Since changing a function on a null set does not change its integral, the result follows. $\square$

Theorem 4 (Monotone Convergence Theorem). *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f, f_1, f_2, \dots$ be $[0, \infty]$ -valued measurable functions on X . Suppose that*

$$f_1 \leq f_2 \leq \cdots$$

and

$$f = \lim_{n \rightarrow \infty} f_n$$

hold almost everywhere. Then

$$\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu.$$

Proof. First assume that $f_n \uparrow f$ pointwise everywhere. By monotonicity of the integral,

$$\int f_1 d\mu \leq \int f_2 d\mu \leq \cdots \leq \int f d\mu.$$

Hence

$$\lim_{n \rightarrow \infty} \int f_n d\mu \leq \int f d\mu.$$

We prove the reverse inequality. For each n , choose an increasing sequence $\{g_{n,k}\}_{k=1}^{\infty}$ of nonnegative simple functions such that

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This is possible by Proposition 2.1.8. Define

$$h_n := \max(g_{1,n}, g_{2,n}, \dots, g_{n,n}).$$

Then $\{h_n\}$ is an increasing sequence of nonnegative simple functions, $h_n \leq f_n$, and

$$h_n \uparrow f.$$

Therefore, by Proposition 2.3.3,

$$\int f d\mu = \lim_{n \rightarrow \infty} \int h_n d\mu.$$

Since $h_n \leq f_n$, monotonicity gives

$$\int h_n d\mu \leq \int f_n d\mu.$$

Thus

$$\int f d\mu = \lim_{n \rightarrow \infty} \int h_n d\mu \leq \lim_{n \rightarrow \infty} \int f_n d\mu.$$

Combining the two inequalities gives

$$\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu.$$

If the assumptions hold only almost everywhere, choose a null set N outside which they hold. Apply the first part to $f_n \mathbf{1}_{N^c}$ and $f \mathbf{1}_{N^c}$. Since changing a function on a null set does not change its integral, the result follows. $\square$

Chapter 3

Chapter 5